

Sales Analytics Star Schema & BI Guide

Comprehensive Architectural Blueprint & SQL Analytics Reference

Prepared from Visual & Technical Documentation | Domain: Business Intelligence & Data Engineering | Target: Executive Analytics

1. Architecture & Entity-Relationship Diagram (ERD)

The Sales Analytics Star Schema is designed for high-performance Online Analytical Processing (OLAP) and business intelligence reporting. It separates business measurements (facts) from contextual descriptive attributes (dimensions), ensuring optimal query execution and simplified analytical modeling.



Figure 1: Entity-Relationship Diagram for Sales Analytics Star Schema showing FactSales and core dimension tables.

Schema Structure Overview

At the center of the architecture is the **FactSales** table, surrounded by core dimension tables linked via foreign key relationships:

- **FactSales (Central Fact Table):** Stores granular transaction metrics including SalesID, CustomerID, Product_id, Store_id, Date_id, Timestamp_id, Promotion_id, Quantity_sold, Unit_price, and Discount_Amount.
- **DimCustomer:** Provides customer demographics including CustomerID, Customer_name, Gender, Age_group, City, and Loyalty_tier.
- **DimProduct:** Categorizes merchandise with Product_id, Product_sku, Product_name, Category, SubCategory, Brand, and Unit_cost.
- **DimStore:** Captures retail location data including Store_id, Store_name, City, State, Country, and Store_type.
- **DimDate:** Calendar dimension containing Date_id, Full_date, Day_Of_Week, Day_number, Month_number, Month_name, Quarter, Year, and Is_Weekend.
- **DimTime:** Time-of-day dimension containing Timestamp_id, Hour, Minute, Second, and Time_Of_Day.

Table Type	Table Name	Key Columns	Primary Analytical Purpose
Fact Table	FactSales	SalesID, CustomerID, Product_id, Store_id, Date_id, Quantity_sold, Unit_price	Measures numeric performance metrics for aggregation
Dimension	DimCustomer	CustomerID, Customer_name, Gender, Age_group, City, Loyalty_tier	Customer segmentation and demographic revenue distribution
Dimension	DimProduct	Product_id, Product_sku, Product_name, Category, SubCategory, Brand	Product category performance, brand, and margin analysis
Dimension	DimStore	Store_id, Store_name, City, State, Country, Store_type	Store location and regional performance tracking
Dimension	DimDate	Date_id, Full_date, Month_name, Quarter, Year, Is_Weekend	Time-series analysis and seasonal trend identification
Dimension	DimTime	Timestamp_id, Hour, Minute, Second, Time_Of_Day	Granular intra-day transaction timing analysis

2. Business Intelligence SQL Queries & Analytics

The following six production-grade SQL queries extract key business intelligence metrics from the star schema, driving executive decision-making and performance monitoring.

Query 1

Total Revenue & Profit Margin Analysis

Calculates total gross revenue, total cost of goods sold (COGS), net profit, and profit margin aggregated by calendar year and month.

```
SELECT
    d.Year,
    d.Month_name,
    SUM(f.Quantity_sold * f.Unit_price) AS TotalRevenue,
    SUM(f.Quantity_sold * p.Unit_cost) AS TotalCost,
    SUM((f.Quantity_sold * f.Unit_price) - (f.Quantity_sold * p.Unit_cost)) AS GrossProfit,
    ROUND((SUM((f.Quantity_sold * f.Unit_price) - (f.Quantity_sold * p.Unit_cost)) / NULLIF(SUM(f.Quantity_sold * f.Unit_price), 0)), 2) AS ProfitMargin
FROM FactSales f
JOIN DimDate d ON f.Date_id = d.Date_id
JOIN DimProduct p ON f.Product_id = p.Product_id
GROUP BY d.Year, d.Month_name
ORDER BY d.Year DESC, d.Month_name DESC;
```

Business Value: Provides finance and leadership teams with month-over-month profitability tracking to evaluate margin health and operational efficiency.

Query 2

Average Order Value (AOV) & Basket Size Analysis

Evaluates customer purchasing behavior by determining average order value and average item quantity per transaction across store regions.

```
SELECT
    s.State,
    COUNT(DISTINCT f.SalesID) AS TotalTransactions,
    SUM(f.Quantity_sold * f.Unit_price) AS RegionRevenue,
    ROUND(AVG(f.Quantity_sold * f.Unit_price), 2) AS AverageOrderValue,
    ROUND(AVG(f.Quantity_sold), 1) AS AvgBasketSize
FROM FactSales f
JOIN DimStore s ON f.Store_id = s.Store_id
GROUP BY s.State
ORDER BY RegionRevenue DESC;
```

Business Value: Guides regional marketing strategies and up-selling initiatives by identifying high-value regional purchasing habits.

Query 3

Product Category & Sub-Category Performance

Ranks merchandise categories by total sales contribution and volume sold to optimize inventory and merchandising.

```
SELECT
    p.Category,
    p.SubCategory,
    SUM(f.Quantity_sold) AS TotalUnitsSold,
    SUM(f.Quantity_sold * f.Unit_price) AS CategoryRevenue,
    RANK() OVER (PARTITION BY p.Category ORDER BY SUM(f.Quantity_sold * f.Unit_price) DESC) AS SubCatRank
FROM FactSales f
JOIN DimProduct p ON f.Product_id = p.Product_id
GROUP BY p.Category, p.SubCategory;
```

Business Value: Essential for supply chain planning, stock allocation, and identifying top-performing product lines.

Query 4

Top Customer Segmentation & Lifetime Value

Identifies top-tier customers by cumulative spend and transaction frequency to support VIP loyalty programs.

```
SELECT
    c.CustomerID,
    c.Customer_name,
    c.Loyalty_tier,
    COUNT(f.SalesID) AS OrderFrequency,
    SUM(f.Quantity_sold * f.Unit_price) AS TotalSpend,
    MAX(d.Full_date) AS LastPurchaseDate
FROM FactSales f
JOIN DimCustomer c ON f.CustomerID = c.CustomerID
JOIN DimDate d ON f.Date_id = d.Date_id
GROUP BY c.CustomerID, c.Customer_name, c.Loyalty_tier
ORDER BY TotalSpend DESC
LIMIT 10;
```

Business Value: Enables targeted retention campaigns and personalized marketing for high-value customer segments.

Query 5

Store Location & Regional Performance Analysis

Evaluates individual store productivity and revenue generation against regional benchmarks.

```
SELECT
    s.Store_id,
    s.Store_name,
    s.State,
    s.Store_type,
    SUM(f.Quantity_sold * f.Unit_price) AS StoreTotalRevenue,
    COUNT(DISTINCT f.CustomerID) AS UniqueCustomers
FROM FactSales f
JOIN DimStore s ON f.Store_id = s.Store_id
GROUP BY s.Store_id, s.Store_name, s.State, s.Store_type
ORDER BY StoreTotalRevenue DESC;
```

Business Value: Assists retail operations in benchmarking store managers and allocating regional marketing budgets.

Query 6

Weekly & Monthly Revenue Trends

Tracks granular time-series revenue progression to spot seasonal spikes, promotional impacts, and sales velocity.

```
SELECT
    d.Year,
    d.Month_name,
    d.Day_Of_Week,
    SUM(f.Quantity_sold * f.Unit_price) AS DailyRevenue,
    AVG(SUM(f.Quantity_sold * f.Unit_price)) OVER (PARTITION BY d.Year, d.Month_name) AS MonthlyAvgDailyRevenue
FROM FactSales f
JOIN DimDate d ON f.Date_id = d.Date_id
GROUP BY d.Year, d.Month_name, d.Day_Of_Week
ORDER BY d.Year, d.Month_name, d.Day_Of_Week;
```

Business Value: Helps operations staff optimize staffing levels and inventory restocking based on day-of-week demand patterns.